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1. Description

GU304A is an engine safety module, which is used for engine safety protection, displaying running status and parameters. The module measures and displays multi-parameters for engine, if warning occurs during engine running, LED on panel will display related warning; if failure occurs, LED on panel will display related failure, module will closes fuel and outputs failure signal at the same time, engine stops. Its main characteristics are as following:

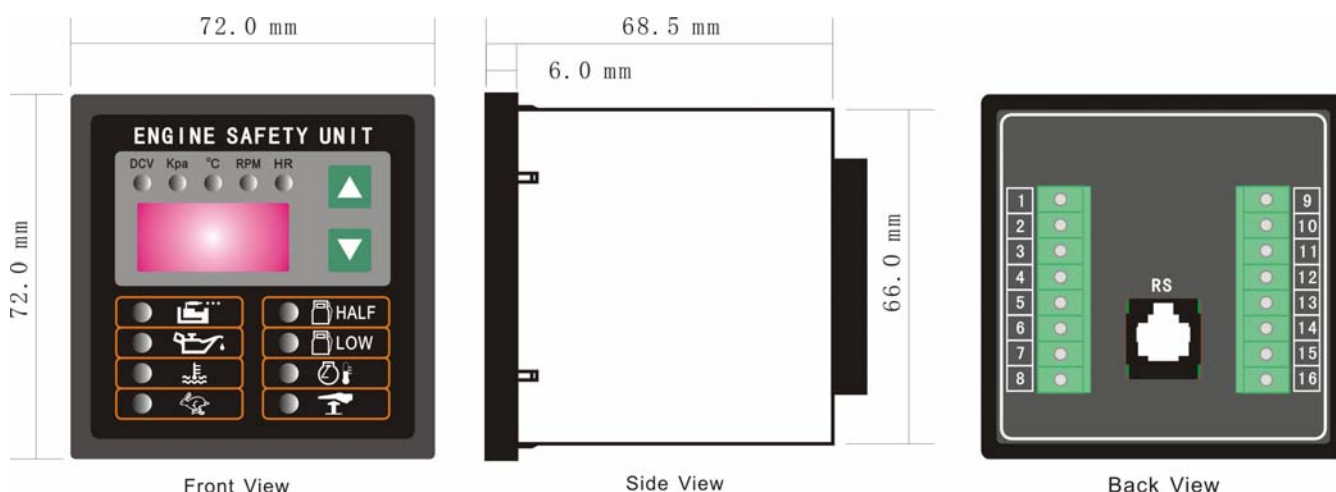
- Adopts MPU as core.
- The protective parameters can be configured neatly by panel buttons and LED.
- Measure and display parameters for engine, such as battery voltage, oil pressure, coolant temperature, rpm and running hours.
- The fuel mode of engine can be set via procedure.
- Inset kinds of P/T sensor for user's choice.
- The controller adopts DIN72 standard enclosure, all connections are connected by pin-like and locked up terminals, easier and more convenient to install, connect, move and replace the device.

This manual is only suitable for GU304A engine controller, user must carefully read this manual first.

2. The Outline Dimensional Drawings and wirings of Controller

2.1 Following Details:

Operation Panel	W72mm×H72mm
Install Hatch	W67mm×H67mm
Thickness	D68.5mm (unconnected)



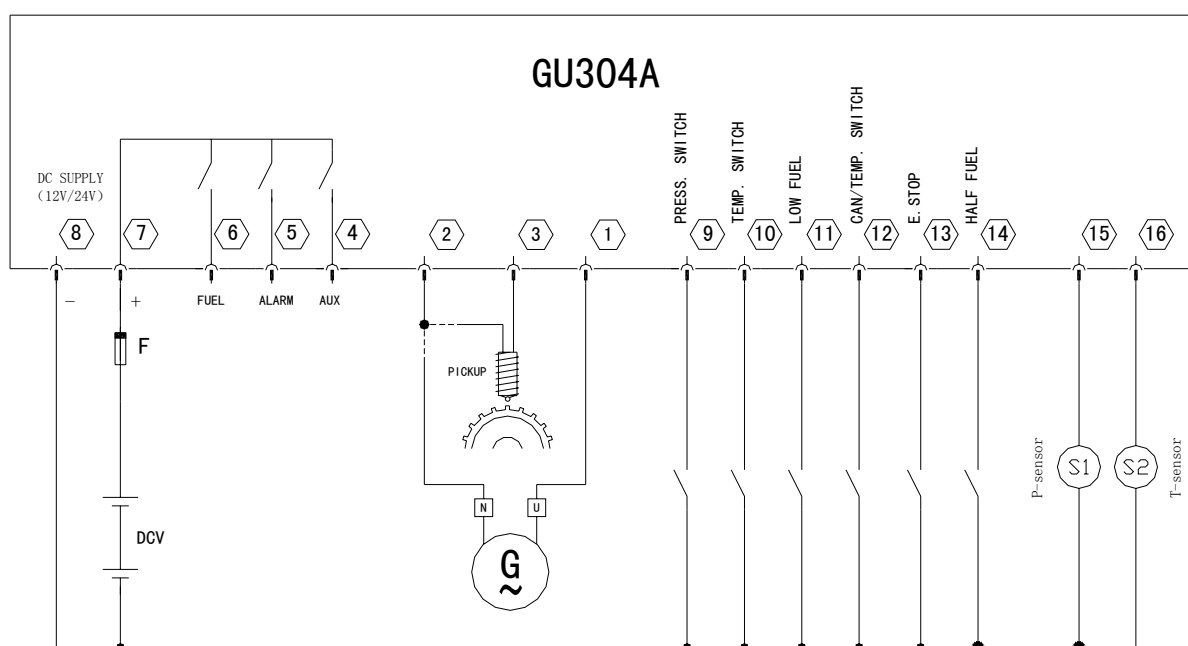
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2.2 Connecting Terminals:

Pin No.	Function Description	Signal	Dim
1	Frequency measure input	50-300VAC (5-70Hz)	1mm ²
2	Frequency com. port		2 core shield line
3	Magnetic sensor signal {+}	1-70VAC (1-10KHz)	
4	Auxiliary output	+B output, 3A/30VDC	1mm ²
5	Alarm (Failure) output	+B output, 3A/30VDC	1mm ²
6	Fuel output	+B output, 3A/30VDC	1mm ²
7	Working power {+}	12V/24V (8-32VDC continued)	1mm ²
8	Working power {-}		1mm ²
9	Low oil pressure signal	low potential is active	1mm ²
10	High temperature signal	low potential is active	1mm ²
11	Low fuel level signal	low potential is active	1mm ²
12	High canopy temperature signal	low potential is active	1mm ²
13	E-stop signal	low potential is active	1mm ²
14	Half fuel level signal	low potential is active	1mm ²
15	Oil pressure detect	resistance sensor (<2KΩ)	1mm ²
16	Coolant temperature detect	resistance sensor (<2KΩ)	1mm ²

NOTE: either frequency measure or speed detect can be selected, do not used simultaneously, connect the common port to Pin No. 2.

2.3 Typical Wiring Diagram:



3. Operation Panel

The whole operation panel includes 3 sections: 4-digit LED, LED indicator and operation buttons.

3.1 LED Displays and the control buttons:

A row of LED on the top of panel respectively display battery voltage (DCV), oil pressure (Kpa), coolant temp (°C), running speed (RPM) and running hours (HR). The subjacent 2 rows of LED respectively display each working status. LED displays each measuring parameter and system menu, and combined with control buttons which provides a friendly operation interface for operator.

Buttons Function Description:

Function Description	Tag
Enter into submenu / modify / confirm modification	 
Scroll up menu / value ascend when continuously press	
Scroll down menu / value descend when continuously press	

3.2 The Descriptions of LED Status:



When safety-on delay counting is over, if module detects that data of oil pressure and coolant temp is normal and no failure shutdown occur, LED lights up, engine is running.



When safety-on delay counting is over, if module detects that oil pressure falls below the low oil pressure alarm value or oil pressure switch signal is active, LED lights up, low oil pressure failure occurs.



When switching on working power source, if module detects that cooling coolant temp exceeds the high coolant temp alarm value or temp switch signal is active, LED lights up, high coolant temp failure occurs.



When switching on working power source, if module detects that running speed exceeds the setting value of speed in module parameters, LED lights up, overspeed failure occurs.



When switching on working power source, if module detects that half fuel level switch signal is active, LED lights up, fuel can on engine stays at half fuel level status.



When switching on working power source, if module detects that low fuel level switch signal is active, LED lights up, fuel can on engine stay at low fuel level status.



When switching on working power source, if module detects that canopy temp switch signal is active, LED lights up, high canopy temperature failure occur.



When switching on working power source, module detects that E-stop switch signal is active, LED lights up, engine stays at the status of emergency stop.

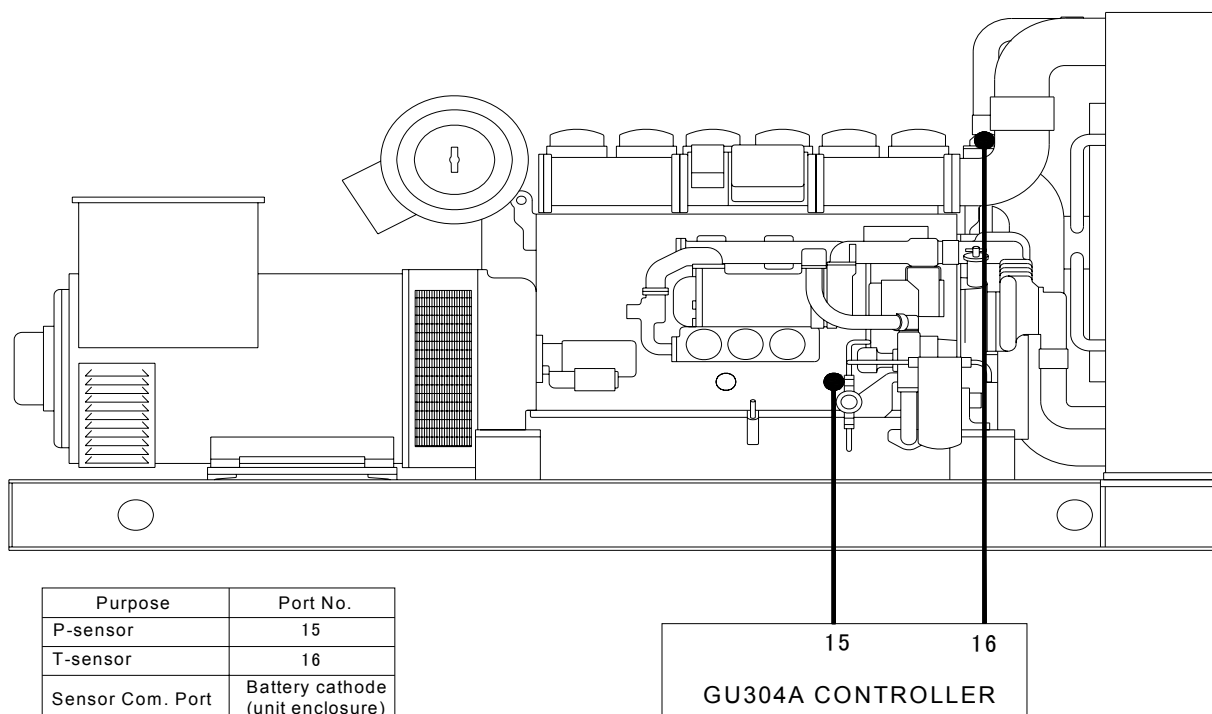
4. Installation Guide

4.1 The dimensional drawing of install hatch on panel, showing as above attached.

The controller is secured by 2 special fittings. The shock-proof equipment must be installed if the enclosure that installed on controller is directly installed on Genset body or other heavy vibrant device.

4.2 Please read above 2.3 Typical Wiring Diagram or attached drawing for wiring connecting.

4.3 The installations of P-sensor and T-sensor for engine:



5. Control and Operation Instruction

5.1 The Process of Start Control:

Operation	Description
When power supply for controller is normal, fuel relay energizes, fuel solenoid valve opens. When controller detects that there is frequency or RPM input, the timer for safety-on delay begins to count. If data of oil pressure and coolant temp of engine unit is normal and no shutdown failure occurs, controller displays running status after counting is over.	

5.2 The start and stop processes of engine, whose fuel solenoid valve is N. O. type (different from N.C. type):

Start control process:

When start running, the fuel relay of controller will not energize, fuel solenoid valve is off power, namely, the electromagnet of solenoid valve do not energize.

Stop control process:

During the stop process, controller outputs the power of fuel solenoid valve, fuel solenoid valve energizes, engine begins to stop. After delay (according to the setting stop delay), cut off the power of fuel solenoid valve.

Other control process is same as engine whose fuel solenoid valve is N. C. type.

6. Measure and Display Data

Battery Voltage (V)	The voltage value is the input source (Battery) voltage value of controller.
Oil pressure (Kpa)	This data is from an external sensor
Coolant Temp (°C)	This data is from an external sensor
Engine Running Speed (RPM)	Signal from engine AC voltage or magnetic sensor
Running Hour (RUN Hour)	The accumulated count of engine each normal running time stored.

7. Warning and Shutdown Alarm

7.1 Warning

(**NOTE:** Warnings are non-critical failure conditions and do not affect the operation of the generator system, they serve for drawing the operators' attention to an undesirable condition and removing it to make sure system continuous running. When warnings occur, the status indicators flash, but failure will not be locked and the unit will not shutdown. Once failure removed, warning indicator will be automatically turned off.)

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Half Fuel Level controller detects the half fuel level switch of engine stay at close status, failure LED “  HALF” lights up.

Low Fuel Level controller detects the low fuel level switch of engine stay at close status, failure LED “  LOW” lights up.

High Canopy Temp controller detects the high temp switch of canopy on engine stay at close status, failure LED “  ” lights up.

7.2 Shutdown Alarm

(NOTE: shutdown failures immediately lock system and stop engine, switching off power is required for reset after failure removed.)

Low Oil Pressure after safety-on delay expired, controller detects that oil pressure value has fallen below the low oil pressure alarm value or oil pressure switch stays at close status, engine stops immediately, failure LED “  ” lights up, failure relay making output.

High Temperature controller detects that engine coolant temp has exceeded the high coolant temp alarm value or temp switch stays at close status, engine stops immediately, failure LED “  ” lights up, failure relay making output.

Overspeed controller detects that engine running speed has exceeded the overspeed alarm value, engine stop immediately, failure LED “  ” lights up, failure relay making output.

Emergency Stop when emergency stop input signal is active, engine stops immediately, failure LED “  ” lights up.

8. Parameters Setting and Menu System

The configuration and modification for parameters adopt gradually increase and decrease, when continuously press increase or decrease button, the unit digit changed one by one, then the tens digit changed after ten unit digit changed, follow this logic, and the hundreds digit changed after tens. Continuously press “▲” and “▼” 2s can enter into setting status, press “▼” can scroll back to view menu, press “▲” and “▼” at the same time to enter into modification, you can press “▲” or “▼” to modify parameter at that time, after finishing modification press “▲” and “▼” at the same time for confirmation. Press “▼” scroll page to **quit** and then press “▲” and “▼” at the same time can quit setting status.

No.	Item	LED	Default	Value Range
01	Fuel mode		0	0~1 / 0 (N.C.) / 1 (N.O.)
02	H-sensor mode		0	0~1 / 0 (frequency) / 1 (rpm)
03	Tooth quantity		120	1~999
04	Pair of poles		2	1~4
05	T-sensor mode		1	0~3 / 0 (not used)
06	P-sensor mode		1	0~3 / 0 (not used)
07	Safety-on delay		60s	0~600s
08	Overspeed Alarm		1600RPM	1~9999 RPM / 9999 (not set)
09	Oil-P low Alarm		140Kpa	0~300 Kpa / 0 (not set)
10	Coolant Alarm		105℃	70~160℃ / 999 (not set)
11	Low battery Alarm		8V	0~25V
12	Auto scroll page		0s	0~10s
13	On line duplication			
14	Quit			

8.1 The Definitions of P-sensor and T-sensor

For P-sensor and T-sensor, according to below table different No. stands for different sensor mode. Bases on the specified No. of practical sensor mode in following table, set the following No. at above No. 05 and No. 06 for the sensor mode in the parameters setting.

No.	P-sensor Mode	T-sensor Mode
0	NOT USE	NOT USE
1	VDO 10 bar	VDO 120 degrees C
2	VDO 5 bar	Datcon high
3	Datcon 10 bar	Four-way

8.2 Static LED Display:

When controller is ready

Operation	Description
If controller is not connected to P-sensor and T-sensor, the corresponding data display "OPEN", LED display: 	 °C  Kpa

When controller set P-sensor and T-sensor are not used

Operation	Description
Controller on standby or running status, the related data all display "----", LED display: 	 °C  Kpa

When controller is normally running

Operation	Description
Controller displays engine status and running information:	RUN 
Press "  or "  can switch display page	 DCV
	 Kpa
	 °C
	 RPM
	 HR

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8.3 Parameters Setting

For Example: (set controller at online program mode)

Operation	Description
Continuously press “▲” and “▼” 2s at the same time, enter into parameters setting menu, then LED display:	
Continuously press “▼” until LED display:	
Continuously press “▲” and “▼” at the same time enter into program mode, controller no any display under this mode and make sure the power supply is normal. Controller will automatically reset and operate over again after the procedure was correctly duplicated using special program software via communication cable, or it will not exit this status until switching off working power.	 

For Example: (Set tooth quantity at 125)

Operation	Description
Continuously press “▲” and “▼” 2s at the same time, enter into parameters setting menu, then LED display:	
Continuously press “▼” until LED display:	
Continuously press “▲” and “▼” 2s at the same time enter into modification, LED display:	
Press “▲” or “▼” to change parameter, change at 125, then LED display:	
After finishing modification press “▲” and “▼” at the same time for confirmation, then continuously press “▼” to QUIT, and then press “▲” and “▼” at the same time for quit.	

For Example: (Set controller overspeed at 1650)

Operation	Description
Continuously press “▲” and “▼” 2s at the same time, enter into parameters setting menu, then LED display:	
Continuously press “▼” until LED display:	
Continuously press “▲” and “▼” 2s at the same time enter into modification, LED display:	
Press “▲” or “▼” to change parameter, change at 1650, then LED display:	
After finishing modification press “▲” and “▼” at the same time for confirmation, then continuously press “▼” to QUIT, and then press “▲” and “▼” at the same time for quit.	

9. Preparation before Start the Controller

- 9.1** Make sure controller is fixed well and its install meets ambient requirement.
- 9.2** Confirm all wiring connections of controller meet electric specification and especially ensure the correct polarity of DC supply source.
- 9.3** We recommend mounting an “Emergency Stop” button externally. The emergency stop input could be connected to N.O. contact of emergency stop push button, and the other contactor point be connected to the battery negative.
- 9.4** Switch on DC working power, make sure the preset parameters meet practical operating conditions, such as fuel mode, T-sensor mode, P-sensor mode and the preset value of kinds of parameters, etc.

10. Technical Parameters

DC Working Source Voltage	12V/24V (8-32VDC continued)
Max. Power Consumption	≤2W
The Measure Range of PICK-UP Voltage Frequency	1-70VAC (1-10000Hz)
The Measure Range of AC Voltage Frequency	50V≤ voltage≤ 300V (5-70Hz)
Control Output Capacity	3A/30VDC
Working Ambient Temperature	-20 to 70℃
Storage Ambient Temperature	-40 to 80℃

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